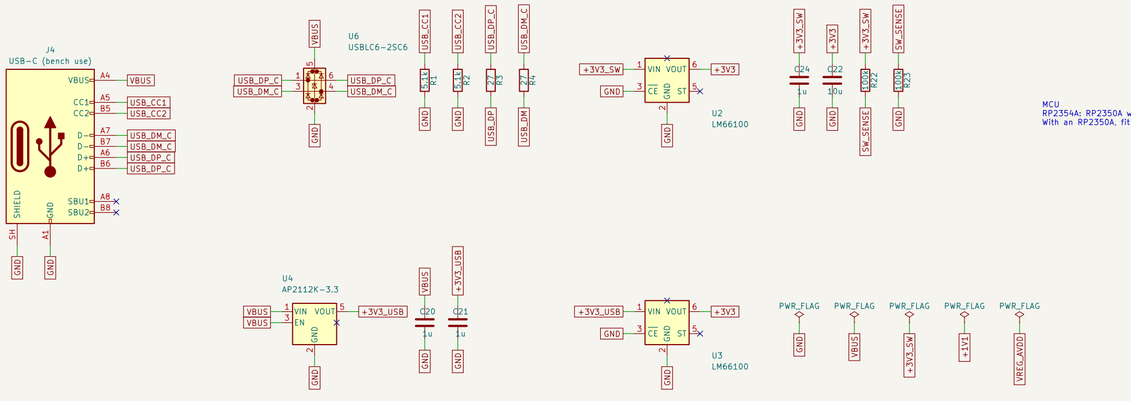
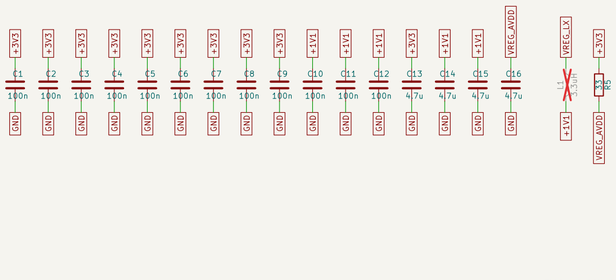
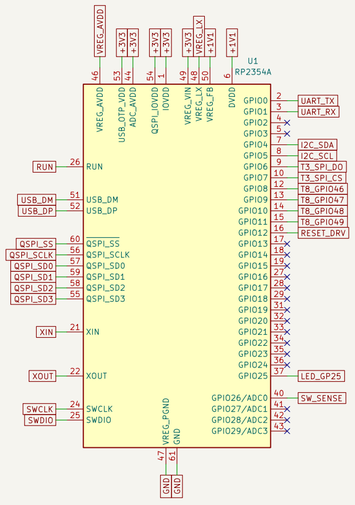
POWER
The switch's 3.3 V (T3/T8 pin 4) and USB's 3.3 V are ORed by two ideal diodes,
so neither back-feeds the other. SW_SENSE tells the firmware whether the switch is on:
until it is, keep every switch-facing pin high-impedance with its input buffer off.



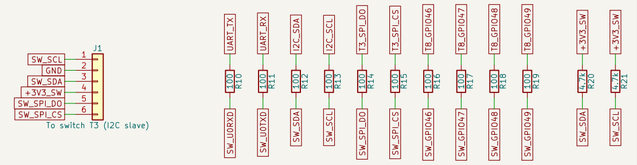
CORE REGULATOR AND DECOUPLING
L1, C13 (CIN), C14 (COUT), R5/C16 (VREG_AVDD filter): lay out as the Pico 2 does (datasheet 6.3.8).
C15 is the second 4.7u at DVDD pin 23. One 100n per IOVDD and DVDD pin, at the pin.



MCU
RP2354A: RP2350A with 2 MiB of flash in the package (same pinout).
With an RP2350A, fit U5 and C19. Never fit U5 with an RP2354A: both would answer on QSPI_SS.

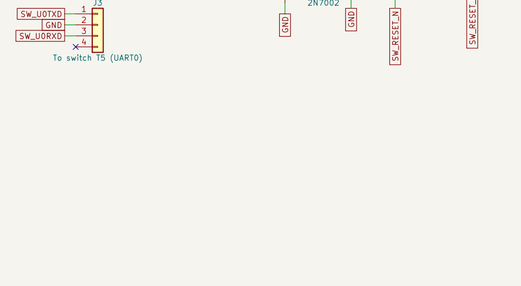


SWITCH HEADERS (J1–J3 are male headers on the board's underside)
100R in series on every switch-facing line: I2C pull-ups to the switch's 3.3 V.
J3 pin 4 (the switch's 3.3 V on T5) is left open.

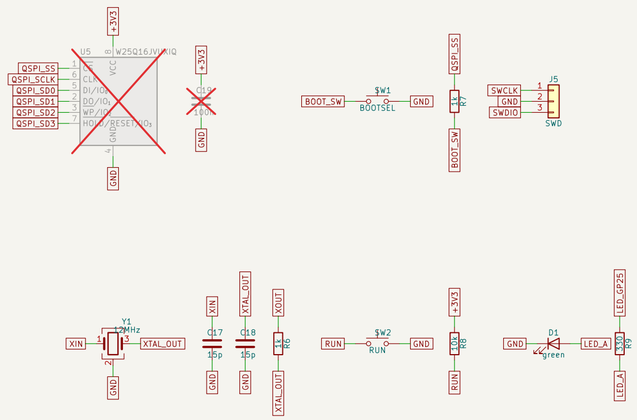




SOC RESET
TP1 goes to D3's ANODE on the switch: the front pad, by the "+".
The banded end is the 3.3 V rail. R25 limits the current if the wire is misplaced.



FLASH, CLOCK, BOOT, DEBUG
U5/C19: DNP with the RP2354A. Keep the QSPI stubs to U5 short.



Generated by modchip/hardware/make_sch.py; not built or tested yet
For SWTG024AS-V2.0 switches: management processor on headers T3, T8 and T5
github.com/wrcrooks/RTLPlayground, modchip/

Sheet: /
File: modchip.kicad_sch

Title: RTLPlayground mod chip (RP2350)

Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.0.5		Id: 1/1